Chapter 1

Su-Schrieffer-Heeger model with multiorbital superconductivity

As a first departure into the search for systems exhibiting topological superconductivity, we begin with a topologically non-trivial system. A canonical topological system is the Su-Schrieffer-Heeger (SSH) model of a spinless, one-dimensional dimer lattice of fermions with staggered hopping. The model obeys a *chiral symmetry*, exhibiting two topological phases. In the topologically non-trivial phase, a pair of edge states emerges at the boundaries, characterised by a *winding number*. We will consider the effect of a spin-triplet, multiorbital attractive interaction within the unit cell, and a Coulomb repulsion to nearest-neighbours outside of the unit cell.

1.1 Su-Schrieffer-Heeger model

The Su-Schrieffer-Heeger model with additional spin degrees of freedom is given by the following Hamiltonian

$$\hat{\mathcal{H}}_{\text{SSH}} = \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow\downarrow\}} \sum_{m\in\{A,B\}} \left[-\mu \hat{c}^{\dagger}_{x,\sigma,m}\hat{c}_{x,\sigma,m}\right] + \sum_{\sigma\in\{\uparrow\downarrow\}} \left[V \hat{c}^{\dagger}_{0,\sigma,m}\hat{c}_{0,\sigma,m}\right] + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow\downarrow\}} \left[-v\hat{c}^{\dagger}_{x,\sigma,A}\hat{c}_{x,\sigma,B} - w\hat{c}^{\dagger}_{x,\sigma,B}\hat{c}_{x+1,\sigma,A} + \text{h.c.}\right], \quad (1.1)$$

where μ is the chemical potential, v is the intracell, interorbital hopping parameter, w is the intercell, interorbital hopping and V is an impurity coupling strength. The model boundary conditions are periodic, and therefore, in the topological state, we expect edge modes to arise around the impurity. In the trivial state, with $v > w$, the unit cell has the strucure displayed in Figure 1.1.

We investigate the phases of the model in Figure 1.2.

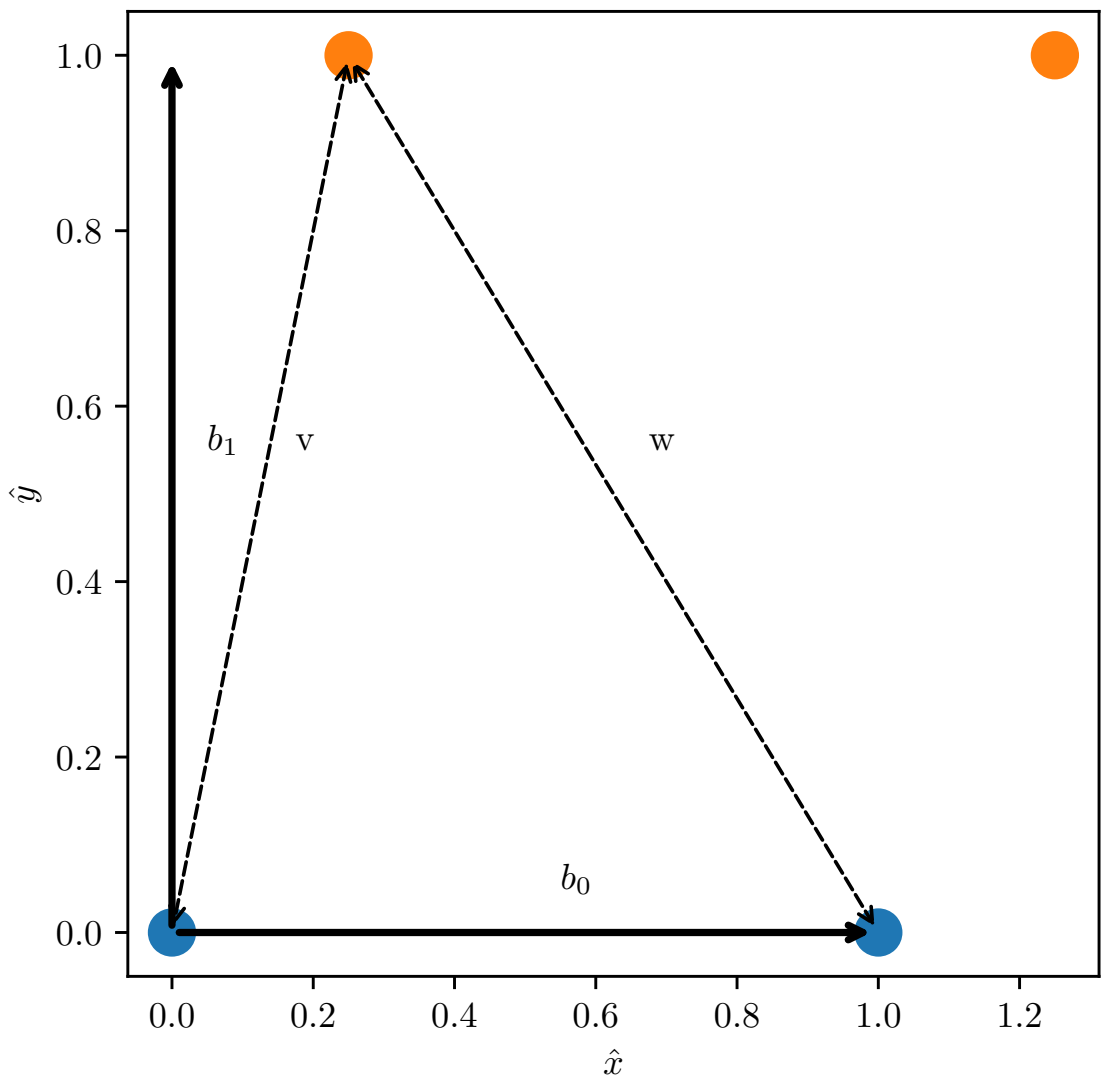

Figure 1.1: The dimer unit cell of the SSH model is depicted, together with hopping parameters w and v, as well as the lattice basis b_0 and b_1. Atoms A and B are colored in blue and orange, respectively.

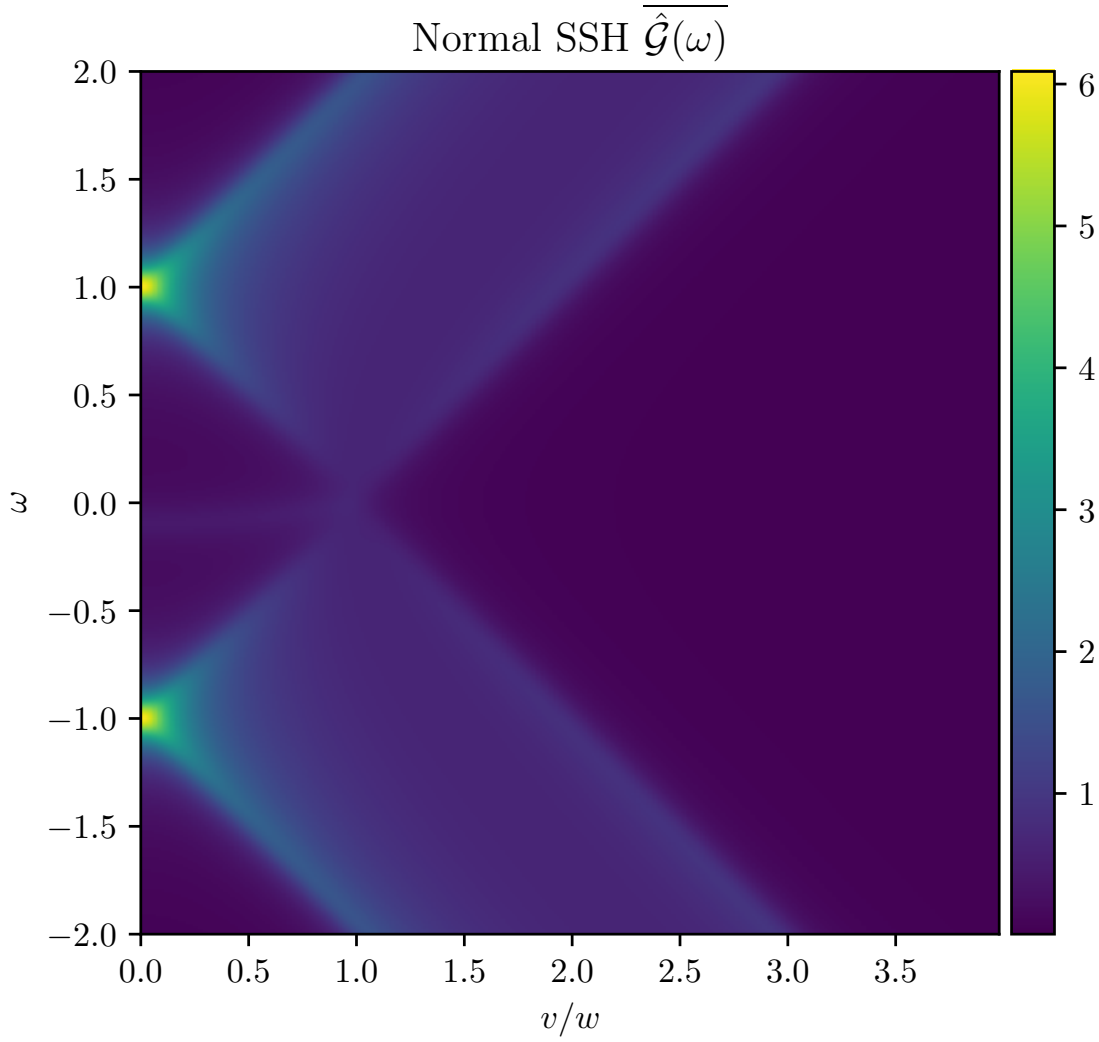

Figure 1.2: Phase diagram of the SSH model in the normal state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. The topological transition occurs at $v/w = 1$ and notice the faint density at $\omega = 0$ below the transition point, due to the edge states. The effect of the chemical potential is to shift the phase boundary, whereas a small impurity shifts the phase boundary on a single site, a large impurity coupling $V = 10.28$ gaps out any modes on the middle site.

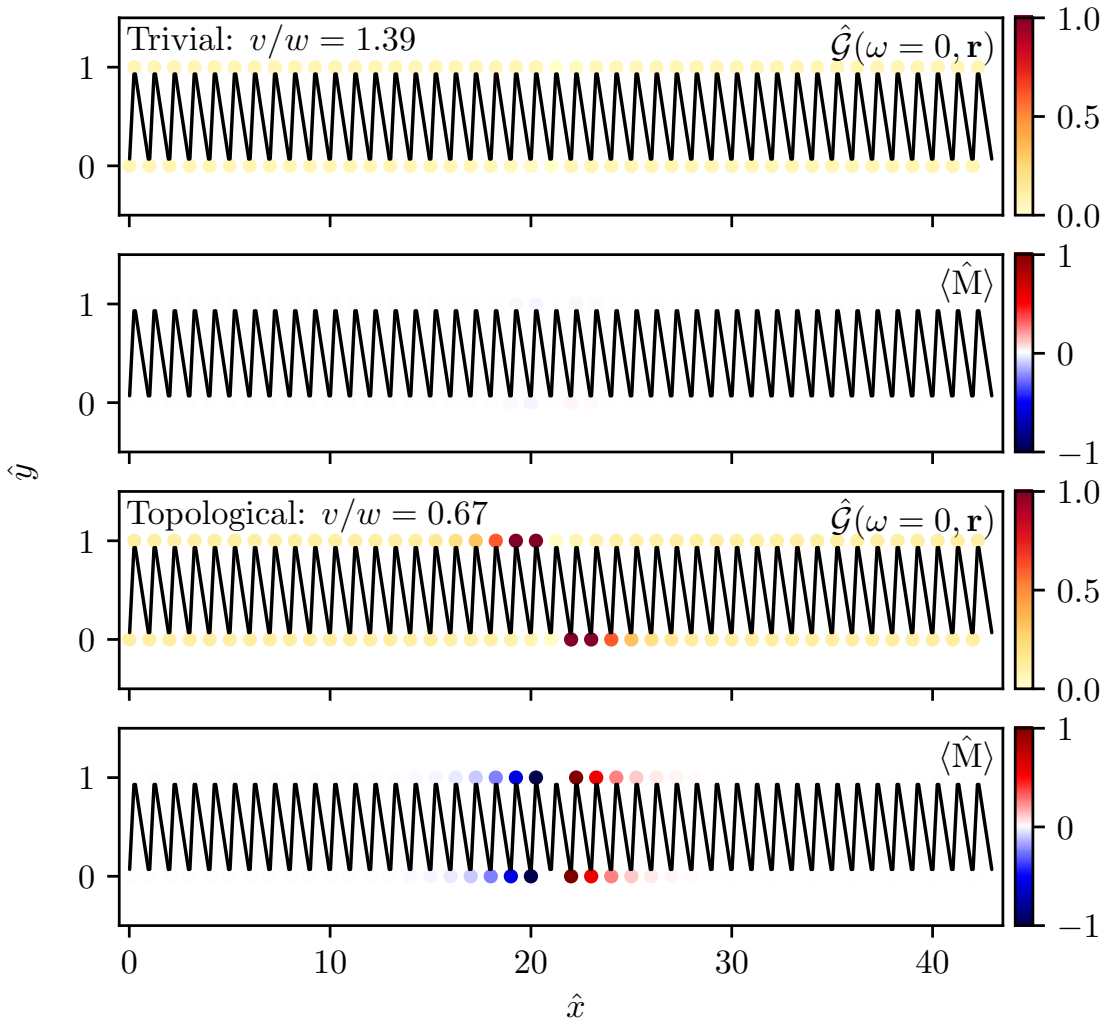

Figure 1.3: The local density of states and magnetism of the SSH model in the normal state with chemical potential $\mu = 0$ and strong impurity coupling strength $V = 10.28$ on the middle site, in order to cut the chain without weak coupling effects. The dark lines represent hopping bonds and the sites are denoted by circles with color corresponding to density. Importantly, we note non-trivial orbital and magnetic characteristics to the edge modes.

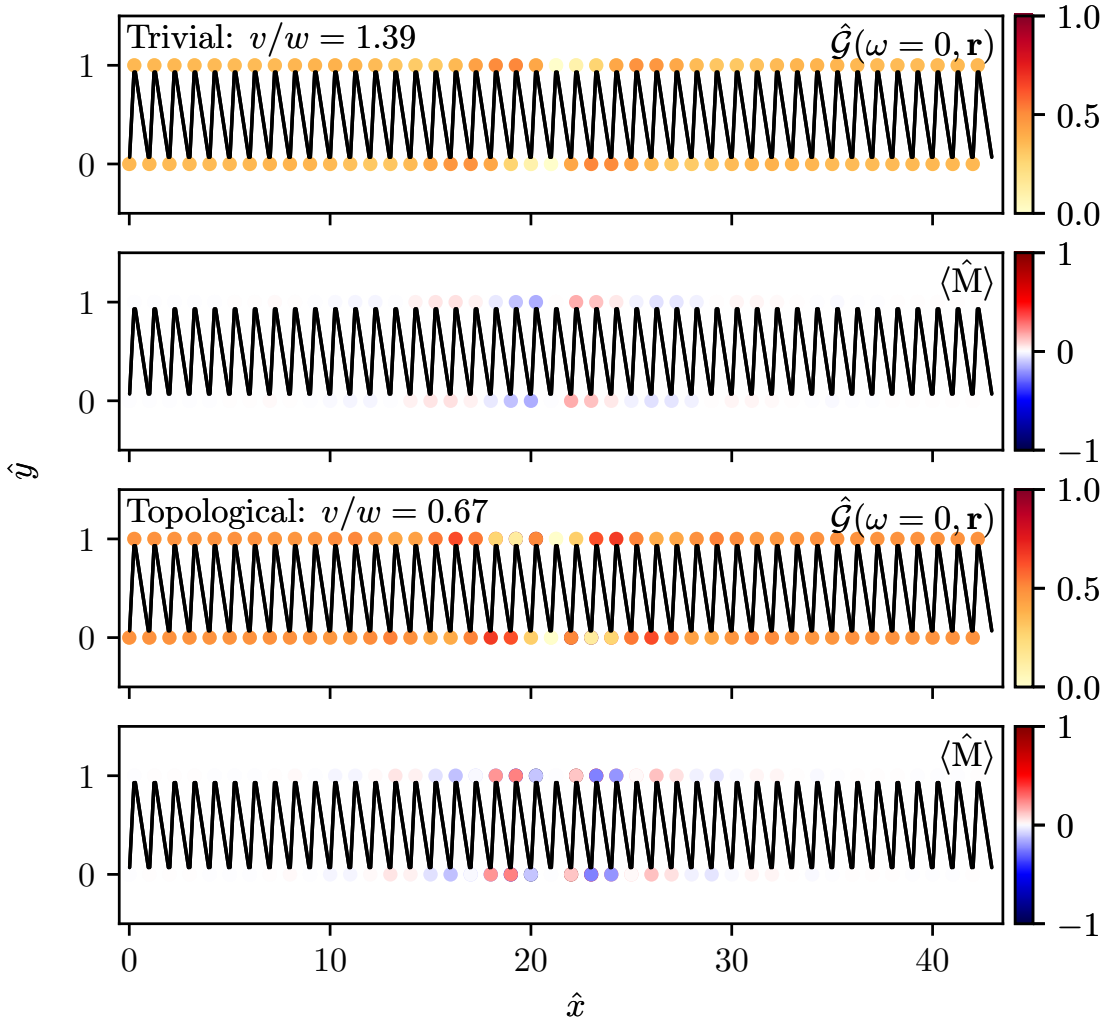

Figure 1.4: The local density of states and magnetism of the SSH model in the normal state with chemical potential $\mu = -0.57$ and strong impurity coupling strength $V = 10.28$ on the middle site. Away from zero chemical potential, we observe a charge density wave in the topological phase.

1.2 Non-unitary spin-triplet Su-Schrieffer-Heeger model

The non-unitary spin-triplet extension of the Su-Schrieffer-Heeger model is given by

$$\hat{\mathcal{H}}_{\text{SSH-triplet}} = \hat{\mathcal{H}}_{\text{SSH}} + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left[-U_v \hat{c}^\dagger_{x,\sigma,A} \hat{c}^\dagger_{x,\sigma,B} \hat{c}_{x,\sigma,B} \hat{c}_{x,\sigma,A} + U_w \hat{c}^\dagger_{x,\sigma,B} \hat{c}^\dagger_{x+1,\sigma,A} \hat{c}_{x+1,\sigma,A} \hat{c}_{x,\sigma,B} \right], \tag{1.2}$$

where U_v is the spin-triplet interaction and U_w is a Coulomb repulsion.

The mean-field solution is

$$\begin{aligned} \hat{\mathcal{H}}^{\text{mf}}_{\text{SSH-triplet}} = \hat{\mathcal{H}}_{\text{SSH}} + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow,\downarrow\}} \Bigg[& \sum_{m\in\{A,B\}} \left[\phi_{x,\sigma,m} \hat{c}^\dagger_{x,\sigma,m} \hat{c}_{x,\sigma,m} \right] \\ & + \Phi^v_{x,\sigma} \hat{c}^\dagger_{x,\sigma,A} \hat{c}_{x,\sigma,B} + \Phi^w_{x,\sigma} \hat{c}^\dagger_{x,\sigma,B} \hat{c}_{x+1,\sigma,A} + \Delta^v_{x,\sigma} \hat{c}^\dagger_{x,\sigma,A} \hat{c}^\dagger_{x,\sigma,B} + \Delta^{v\dagger}_{x,\sigma} \hat{c}_{x,\sigma,B} \hat{c}_{x,\sigma,A} \\ & + \Delta^w_{x,\sigma} \hat{c}^\dagger_{x,\sigma,B} \hat{c}^\dagger_{x+1,\sigma,A} + \Delta^{w\dagger}_{x,\sigma} \hat{c}_{x+1,\sigma,A} \hat{c}_{x,\sigma,B} \Bigg], \end{aligned} \tag{1.3}$$

where the Hartree, Fock and Gorkov fields are defined as

$$\phi_{x,\sigma,m} = (U_v - U_w) \hat{c}^\dagger_{x,\sigma,m} \hat{c}_{x,\sigma,m} \tag{1.4}$$

$$\Phi^v_{x,\sigma} = -U_v \hat{c}^\dagger_{x,\sigma,A} \hat{c}_{x,\sigma,B} \tag{1.5}$$

$$\Phi^w_{x,\sigma} = \hat{U}_w c^\dagger_{x,\sigma,B} \hat{c}_{x+1,\sigma,A} \tag{1.6}$$

$$\Delta^v_{x,\sigma} = -U_v \hat{c}_{x,\sigma,B} \hat{c}_{x,\sigma,A} \tag{1.7}$$

$$\Delta^{v\dagger}_{x,\sigma} = U_w \hat{c}^\dagger_{x,\sigma,A} \hat{c}^\dagger_{x,\sigma,B} \tag{1.8}$$

$$\Delta^v_{x,\sigma} = -U_v \hat{c}_{x+1,\sigma,A} \hat{c}_{x,\sigma,B} \tag{1.9}$$

$$\Delta^{w\dagger}_{x,\sigma} = U_w \hat{c}^\dagger_{x,\sigma,B} \hat{c}^\dagger_{x+1,\sigma,A} \tag{1.10}$$